SelfMTL: Self-Supervised Meta-Transfer Learning via Contrastive Representation for Hyperspectral Target Detection

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***Abstract*— Hyperspectral target detection (HTD) is an approach to identify targets of interest in a scene by utilizing prior target spectra. Existing deep learning-based HTD methods usually need to generate a large number of samples for network training, and these generated samples often suffer from distortion. In addition, most of the methods can only be applied to a single scene. To address these issues, this study proposes a self-supervised meta-transfer learning (SelfMTL) method to improve the generalization ability and adaptability of the model through contrastive representation. First, labeled source data, which contains rich feature information, is utilized to train the global–local spectral contrastive learning (GLSL) module by randomly constructing positive and negative pairs from different land covers for the classification task, aiming to effectively discriminate the similarities and differences between spectra. Then, a small sample (only one target–background pair) fine-tuning is utilized to transfer the pretrained GLSL to different target detection (TD) tasks. Finally, a novel adaptive spatial–spectral enhancement (ASSE) module is proposed, which takes into account the joint learning constraints of spatial and spectral information to obtain the final detection result map. The experimental results on four real hyperspectral images (HSIs) datasets verify the superiority of SelfMTL in comparison to many classical and SOTA HTD methods. The codes are available at https://github.com/ShissHAN/SelfMTL.**

***Index Terms*— Contrastive learning, hyperspectral target detection (HTD), meta-learning, self-supervised learning, transfer learning.**

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I. Introduction

HYPERSPECTRAL images (HSIs) typically contain hundreds or even thousands of spectral bands with extremely high spectral resolution. This high resolution allows HSIs to accurately distinguish the characteristics of an object or scene between bands, providing a unique advantage for hyperspectral target detection (HTD) missions. The goal of HTD is to use prior target spectra to identify and localize a target of interest [1], [2]. This technique capitalizes on the differences in the spectral response of target objects under different bands and is highly relevant in current Earth observation and remote sensing tasks. Therefore, HTD has a wide range of applications in the fields of military reconnaissance [3], medical diagnosis [4], and environmental monitoring [5].

In the early years, researchers have introduced numerous HTD approaches, typically including constrained energy minimization (CEM) [6] based on finite alluvial response filters and orthogonal subspace projection (OSP) [7]. To further enhance detection efficiency, some researchers proposed a series of variants of CEM algorithms, such as the hierarchical CEM (HCEM) detector [8], the ensemble-based CEM (E-CEM) [9], and so on. However, these methods are difficult to utilize the nonlinear characteristics of spectra and are less adaptable to complex detection scenarios. Therefore, to address the shortcoming, many researchers have proposed target detectors based on kernel methods, such as kernel orthogonal subspace projection (KOSP) [10], kernel-based CEM (KCEM) [11], and kernel-matched subspace detectors (KMSDs) [12]. Kernel-based HTD methods depend on the assumption that the data become linearly separable once transformed into a high-dimensional kernel feature space. To avoid the need for explicit assumptions about the statistical distribution of the data, representation-based HTD methods have been developed, such as the sparsity-based TD (STD) [13], the combined sparse and cooperative representation detector (CSCR) [14], the hybrid sparsity and distance-based discrimination detector (HSDD) [15], and the learning single spectral abundance (LSSA)-based subpixel TD [16]. In addition, many other machine learning methods have been introduced, including metric learning [17], tree-structured encoding [18], and so on. However, traditional machine learning models usually extract shallow features, which are more limited in the case

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where the target and the background are complex nonlinear differentiable, and cannot obtain ideal detection results.

In recent years, due to its powerful nonlinear representation capabilities, deep learning has achieved breakthrough progress in various fields of hyperspectral image processing, such as change detection [19], [20], land-cover classification [21], [22], [23], unmixing [24], and anomaly detection [25], [26], [27]. For HTD, prior information is usually scarce, and the class imbalance between targets and backgrounds is significant, making it difficult to utilize deep learning networks for this task. Therefore, researchers have proposed a series of frameworks based on deep learning-based HTD algorithms. For example, Zhang et al. [28] proposed a deep CNN-based HTD algorithm, HTD-Net. In addition, in the literature [29], to reduce the loss of spectral information, Zhu et al. [29] proposed a two-stream CNN (TSCNTD). Furthermore, Gao et al. investigated the use of an auxiliary generative adversarial network (GAN) [30] to produce supplementary target and background samples. Furthermore, interpretable representation networks-based HTD algorithms [31], deep metric learning-based HTD algorithms [32], and lightweight convolutional neural network-based HTD algorithms [33] have also exhibited competitive performances.

The aforementioned algorithms are usually trained and tested on the same image scenes and rely on a small number of target priors to synthesize training samples, which requires a significant amount of computational time. Additionally, training and testing on a single image results in poor generalization of these algorithms, making it difficult to transfer the learned features to new datasets for TD. This greatly limits the application scenarios of these algorithms. To alleviate the aforementioned problems, some algorithms based on few-shot learning have been proposed. Shi et al. [34] proposed a semi-supervised domain adaptive few-shot learning (SDAFL)-based detector to transfer pretrained models to the target domain. In addition to this, some algorithms based on self-supervised learning have also been proposed to alleviate the aforementioned problems. For example, Wang et al. [35] proposed a self-supervised spectral-level contrastive learning-based HTD method (SCLHTD) to learn spectral similarity. In addition, Zhang et al. [36] proposed a self-supervised HTD method based on deep clustering (SSDCTD) and Chen et al. [37] proposed a self-supervised generative learning algorithm (GSSL), which also achieves competitive performance. However, the above methods still require time to screen or generate samples for fine-tuning specific HTD tasks.

To avoid time-consuming data generation processes and significant discrepancies between generated and real data, this article proposes a new self-supervised meta-transfer learning (SelfMTL) HTD network. To explore much more information on hyperspectral images, we designed a novel network based on global–local vision and spatial frequency fusion modules (SFFM). The global and local visual models explore the nonlocal and local information of hyperspectral image patches based on the transformer and CNN, respectively. Based on the new network, the SelfMTL method employs a new meta-transfer strategy to train the network and transfer it to different HTD tasks. The proposed SelfMTL consists of two individual components: SelfMTL and adaptive spatial–spectral enhancement (ASSE) detector. First, during the meta-training phase, we utilize labeled HSI classification data to randomly form multiple positive and negative sample pairs from different land covers to train the model for effectively discriminating the similarities and differences between spectra and enhancing the model's sensitivity to HSI spectral changes. Next, the GLSL module as well as the max distance triplet (MDTriplet) loss are utilized to train the model for efficiently discriminating the spectral differences and transfer the pretrained model to different TD tasks, using one sample from each target background for fine-tuning. Finally, an ASSE model is adopted, which jointly learns the spatial and spectral information constraints to obtain the final detection result.

The main contributions of our research are summarized as follows.

1) To analyze the global and local information of the spectra more efficiently, we propose a GLSL network that can effectively distinguish the dissimilarity between spectra by extracting the local and nonlocal information using local visual (LV), global visual (GV), and SFFM modules.
2) To address the challenges of limited training samples and the limitations of single-scenario applications, we propose a SelfMTL framework that reduces the dependence on labeled training samples while quickly adapting to new TD tasks.
3) To efficiently fuse HSI spatial and spectral feature information, the ASSE module is designed to realize the learning of joint spatial–spectral constraints, which enhances the suppression ability of backgrounds to improve the discriminating performance of targets.

The remainder of the article is organized as follows. In Section II, the proposed SelfMTL method is introduced in detail. Then, the experiments and the corresponding analysis are presented in Section III. Finally, conclusions are drawn in IV.

II. Method

In this section, the proposed method is introduced in detail. We first present the overall architecture. Then, the three key components are elaborated, including global–local spectral contrastive learning (GLSL), meta-transfer learning, and ASSE.

A. Overall Architecture

The overall framework of the proposed method is shown in Fig. 1. As shown in the figure, to alleviate the problem of insufficient HTD samples for training, we first pretrain the proposed GLSL network using an open-source labeled hyperspectral image dataset, which is always used for classification. We randomly construct positive sample pairs from the same category and negative sample pairs from different categories. Subsequently, we adopt the MDTriplet loss and input the constructed positive and negative sample pairs into the GLSL module for spectral-level contrastive learning. In GLSL, we have used a network architecture that we propose, and the overall GLSL network architecture is shown in Fig. 2,

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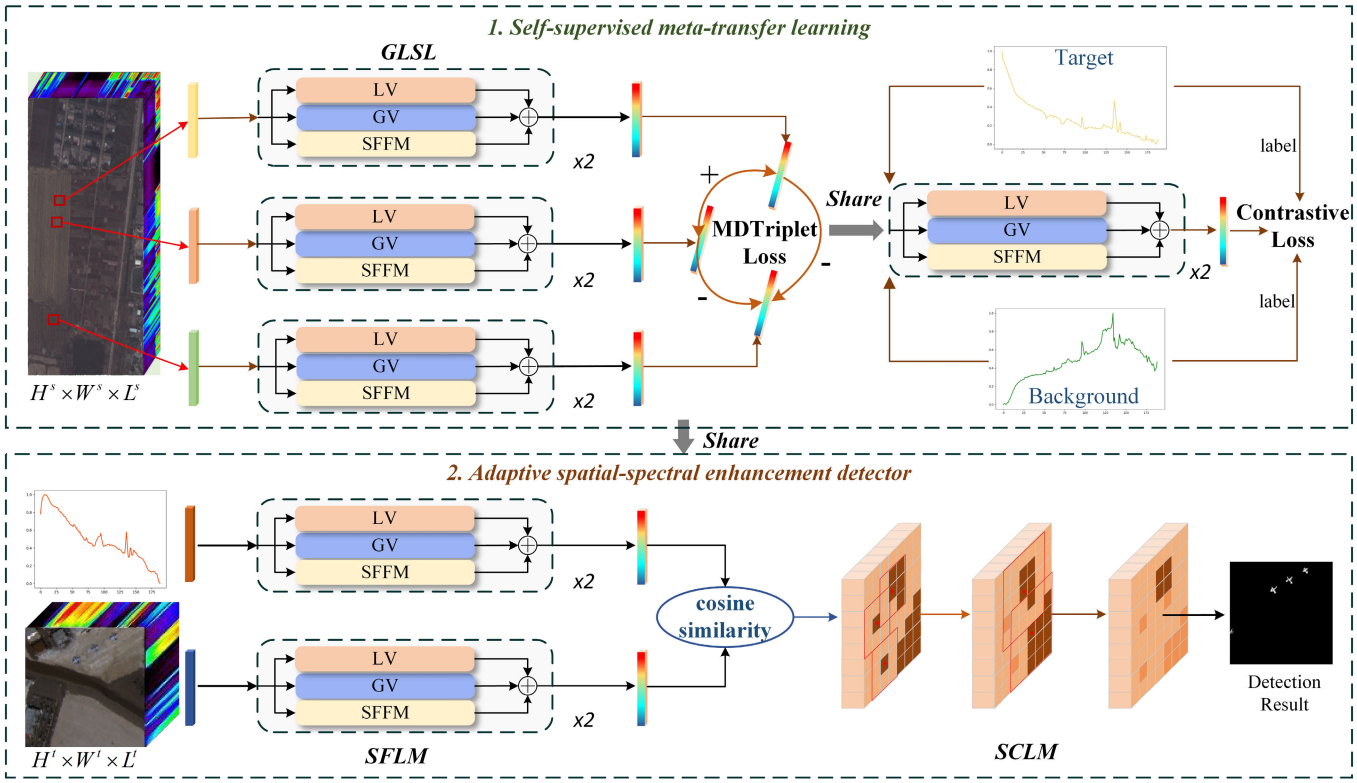


Fig. 1. Flowchart of the proposed SelfMTL method. It mainly contains two modules: SelfMTL and ASSE detector.

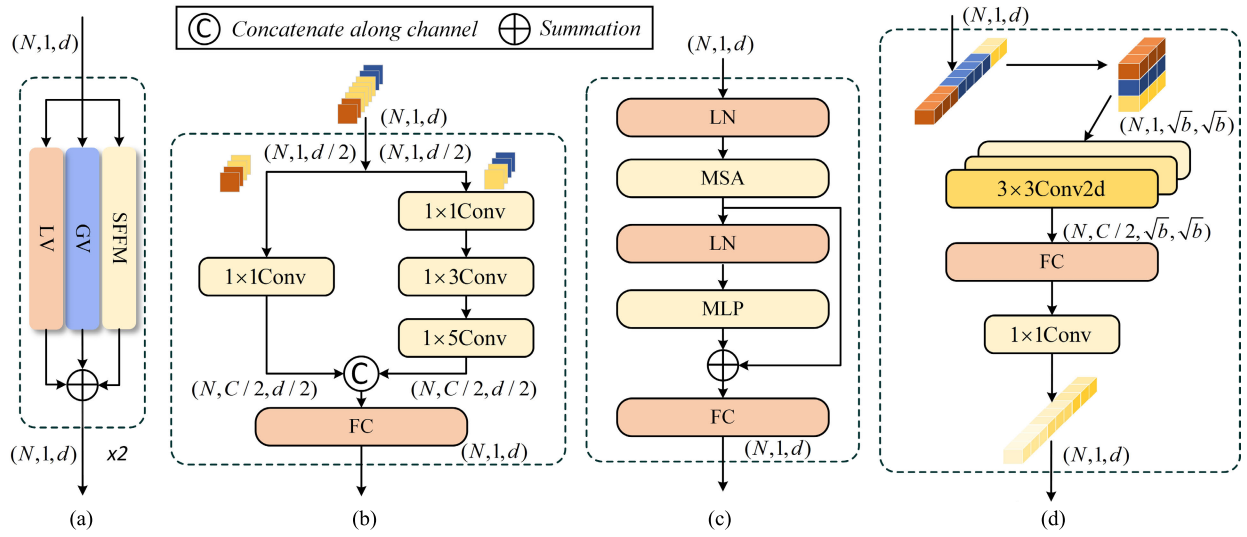


Fig. 2. Illustration of GLSL in the proposed SelfMTL. (a) GSLS. (b) LV. (c) GV. (d) SFFM.

which contains a local and global learning model as well as the SFFM model, which transforms the 1-D vector into the 2-D tensor and performs the 2-D CNN to explore nonlocal information of different bands. Moreover, to effectively transfer the GLSL module, which is capable of distinguishing significant spectral differences. For various TD tasks, we used positive and negative sample spectra from the target and background for model parameter fine-tuning. During the testing phase, we input the prior target spectra into the GLSL module to obtain feature-enhanced spectral bundle representations. We then calculate the similarity between these spectral bundle features and use them to form a spectral similarity map. Finally, to fully exploit the spatial information of the target, an ASSE module is used to achieve joint spatial–spectral constraint learning to optimize the final classification map. To better elucidate the proposed algorithm, Sections II-B–II-D present each step in the order of GLSL, meta-transfer learning, and ASSE.

B. Global–Local Spectral Contrastive Learning

Hyperspectral data typically covers dozens or even hundreds of spectral bands, containing a large amount of redundant

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information and high-dimensional features. To detect the target area based on prior spectral information, we need to identify whether the spectra are similar to the prior spectra. To effectively learn the similarities among spectra of the same class and the differences among spectra of different classes, we introduced an innovative GLSL module. As shown in Fig. 1, the inputs of GLSL are $D_a \in \mathbb{R}^{d\times 1}$, $D_p \in \mathbb{R}^{d\times 1}$, and $D_n \in \mathbb{R}^{d\times 1}$, representing two same-class spectra and one different-class spectra, and the outputs are the feature-enhanced spectral beams D'_a, D'_p, and D'_n. To effectively learn the similarities among spectra of the same class and the differences among spectra of different classes, we introduced an innovative GLSL module. Building upon GLSL, we utilized a Siamese network structure to adequately simulate the differences between target and background spectra for contrastive learning. Specifically, the GLSL module consists of three branches: LV, GV, and SFFM.

1) LV: In spectral analysis, subtle differences between adjacent bands often contain important information. Convolution can effectively extract features from these adjacent bands, helping the model to understand the interactions between different bands. Specifically, as illustrated in Fig. 2(b), we designed the LV module that partitions the input spectral data into two parts, $F_1 \in \mathbb{R}^{N\times 1\times d/2}$ and $F_2 \in \mathbb{R}^{N\times 1\times d/2}$, where N denotes the number of sample batches and d denotes the number of bands, which are processed separately. F_1 utilizes a 1×1 convolution to adjust the dimension of the features and serves as a nonlinear transformation to introduce additional nonlinearity. F_2 extracts multilevel local features from the spectral data by applying different sizes of convolution kernels in a layered manner. Finally, the features obtained from parallel processing are fused, and a fully connected layer is employed to map the extracted features into a lower-dimensional space, resulting in the feature map L

$$L = \mathrm{FC}(\mathrm{Concat}(\mathrm{Conv}_{1\times 1}(F_1), \mathrm{Conv}_{1\times 5}(\mathrm{Conv}_{1\times 3}(\mathrm{Conv}_{1\times 1}(F_2))))) \quad (1)$$

where $\mathrm{Concat}(\cdot)$ refers to the concatenation operation and $\mathrm{Conv}_{1\times 5}$ represents a 1×5 1-D convolution.

2) GV: Although the convolution operation can preserve the local details of the spectra to a certain extent, important global information may be overlooked for the tens or even hundreds of bands that spectral bands usually contain. Therefore, in the GV module, we adopt the transformer structure to establish global relationships and distant dependencies through the self-attention mechanism. Specifically, as shown in Fig. 2(c), the input spectral data F is sequentially passed through the normalization, multihead self-attention (MSA), multilayer perceptron (MLP) layers, and FC layers, thus extracting spectral features from multiple neighboring bands and fusing them through residual connectivity, which reduces the loss of information from shallow to deep layers. In general, the GV module extracts and integrates global features using the MSA mechanism and the MLP component. Through the combination of these components, the module can capture the complex dependencies between different frequency bands when processing hyperspectral data, allowing the learning of global features, which is defined as G

$$G = \mathrm{FC}(\mathrm{MLP}(\mathrm{LN}(\mathrm{MSA}(\mathrm{LN}(F))) + \mathrm{MSA}(\mathrm{LN}(F)))). \quad (2)$$

3) SFFM: To further improve the quality of the feature representation, we designed the SFFM module to integrate the local details of the spectra with the global contextual information. Therefore, in SFFM, we reshape the 1-D input F into two dimensions and perform feature extraction by 2-D convolution, which is beneficial for obtaining local correlations in the spectral data and learning the potential relationships between originally nonadjacent spectral bands. Specifically, as shown in Fig. 2(d), the 1-D spectra $F \in \mathbb{R}^{N\times 1\times d}$ is transformed into $F_{\mathrm{pad}} \in \mathbb{R}^{N\times 1\times\sqrt{b}\times\sqrt{b}}$, with b being the nearest open square, followed by three 3×3 2-D convolutional layers for spatial feature learning, with a ReLU activation function after each step. The 2-D feature maps are then spread to 1-D vectors, and the spread feature vectors are converted back to 1-D data using a fully connected layer. Finally, a 1-D convolutional layer is used to perform the final feature extraction on the converted 1-D data. Through the above series of operations, the SFFM can effectively fuse the local and nonlocal features of the input spectra, thus the strong expressiveness of the features is obtained as a feature map S

$$S = \mathrm{Conv}_{1\times 1}\Big(\mathrm{reshape}\Big(\mathrm{FC}\Big(\mathrm{Flatten}\Big(\mathrm{Conv}_{3\times 3}^{(k)} \times \big(\mathrm{reshape}\big(F_{\mathrm{pad}}\big)\big)\Big)\Big), d\Big) \quad (3)$$

where k denotes three consecutive passes through the convolutional layer.

The three proposed feature extraction modules consider both global and local feature capture. Finally, we map these three different features to a lower dimensional space for feature fusion to obtain the fused feature map H and further learn deeper spectral detail features by using the GLSL module twice

$$H = L + G + S. \quad (4)$$

C. Meta-Transfer Learning

Hyperspectral classification data enable comparative learning to deeply mine and distinguish multiple nuances because they contain rich classes and detailed spectral information, while HTD usually focuses on only two classes of spectra, the target, and the background, together with a very small percentage of the target spectra, which limits the diversity and depth of its training. Therefore, to solve the above problems we utilize the categorized data to train GLSL with a strong ability to discriminate different classes of spectra. How to effectively transfer the GLSL module with the ability to discriminate different class spectra obtained in the pretraining stage for the HTD task is an important issue. In the proposed HTD algorithm, we adopt the meta-transfer learning scheme. The pretraining phase of learning by spectral comparison, where the model is pretrained on large-scale data, learns generic feature representations. These features are then used in a small-sample learning phase to adapt to a specific task or

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dataset. In addition, considering the scarcity of target spectra, only one target sample and one background sample are utilized for fine-tuning the model. As shown in Fig. 1, it is mainly divided into two parts: the spectral-contrast learning pretraining and the small-sample fine-tuning.

1) Spectral-Contrast Learning Pretraining: The spectral contrast learning pretraining module aims to train a network with strong spectral discrimination capabilities using a large amount of labeled open-source hyperspectral image datasets, which can alleviate the problem of insufficient sample size. First, we construct a large number of positive and negative sample pairs using data from labeled samples for training. Second, to better extract local and nonlocal features of HSIs, we propose a new network structure. Finally, to make good use of these constructed positive and negative sample pairs for training, we use a triplet loss to train the network. Since the GLSL module has already been introduced, we focus on the sample construction (training data) and loss function (MDTriplet loss) modules.

a) Training data: Since the proportion of target and background data samples in HTD datasets is extremely unbalanced, to solve this problem, we use a labeled hyperspectral classification dataset with a more balanced distribution for pretraining. A given hyperspectral classification dataset can be labeled as $D_s \in \mathbb{R}^{H^s \times W^s \times L^s}$ and contains C^s classes. To enhance the model's ability to perceive spectral variations, we perform spectral-level contrast learning by constructing positive and negative sample pairs. Positive sample pairs are spectra from the same category, and the model can better capture the small differences between spectra of the same category, thus improving generalizability. Negative sample pairs are spectra from different categories, which help the model to better distinguish between different categories, thus reducing the misclassification rate and improving overall classification performance. Using the idea of meta-representation, two types of spectra are randomly selected at each epoch to form a set of positive and negative sample pairs, increasing the diversity of tasks and thus increasing the generalization ability of the model.

b) MDTriplet loss: Triplet loss is a loss function for learning contrastive feature representations, which is commonly used to train contrastive learning models. The goal of triplet loss is to make the distance between anchor points and positive samples as small as possible and the distance between anchor points and negative samples as large as possible. The specific loss function for a general triplet loss can be defined as follows (using the Euclidean distance as an example):

$$\mathcal{L}_{\text{triplet}} = \sum_i \left[\|F(A_i) - F(P_i)\|_2^2 - \|F(A_i) - F(G_i)\|_2^2 + \alpha\right]_+ \quad (5)$$

where $\|\cdot\|_2$ denotes the Euclidean norm and $F(A_i)$, $F(P_i)$, and $F(G_i)$ denote the feature extraction function applied to the anchor sample A_i, positive sample P_i, and negative sample G_i, respectively. α is the margin parameter and $[x]_+ = \max(x, 0)$ is the operation that takes a positive value to ensure that the loss is nonnegative.

Although the general triplet loss plays a role in the classification task, if there are situations where the positive and negative samples are very close to each other, the model may over-optimize these samples. Therefore, inspired by the ISIA loss [38] function, the MDtriplet loss is designed on this basis considering the distance between positive and negative samples at the same time. It aims to go for a more comprehensive evaluation of the model's performance by comparing the distance between the anchor point and the negative samples, as well as the distance between the positive samples and the negative samples, to improve the model's robustness and generalization ability. This MDtriplet loss is described as follows:

$$\begin{aligned}\mathcal{L}_{\text{MDtriplet}} &= \frac{1}{N}\sum_{i=1}^{N}\Big[\max\Big(\|F(A_i) - F(P_i)\|_2^2 \\ &\quad - \max\Big(\|F(A_i) - F(G_i)\|_2^2, \|F(P_i) - F(G_i)\|_2^2\Big) + \alpha, 0\Big)\Big] \end{aligned} \quad (6)$$

where N is the number of samples in the batch.

2) Small-Sample Fine-Tuning: As shown in Fig. 1, the fine-tuning of the small sample is divided mainly into two parts: the Siamese neural network and contrast loss.

The Siamese neural network performs well with a small amount of labeled data and is particularly suitable for contrast learning tasks. Therefore, we adopt the Siamese neural network structure, embed the already trained GLSL module into the network structure, and fine-tune the whole GLSL module using small-sample pairs. Specifically, a target and a background sample in the HTD dataset are randomly selected, and we input the target or the background sample with the target prior into the network for spectral feature enhancement and compute the similarity between the enhanced spectra and the prior knowledge.

Contrastive loss is used to measure the similarity between pairs of samples and is commonly used to train deep learning models for similarity learning. It optimizes the model by minimizing the distance between pairs of similar samples and maximizing the distance between pairs of dissimilar samples. More importantly, the contrast loss function can effectively learn useful features with limited samples and reduce the need for large amounts of labeled data. Therefore, we use contrastive loss to minimize the distance between the positive samples and the target prior and to maximize the distance between the negative samples and the target prior. Specifically, let the embedding vectors of the sample pairs (x_1, x_2) be z_1 and z_2, where z_i is the embedding representation of the output from the network. Then, we compute the similarity between the positive and negative samples and the target prior

$$d = \|z_1 - z_2\|_2^2. \quad (7)$$

Then, the mathematical formula for the contrastive loss function is

$$\mathcal{L}_c = \frac{1}{N}\sum_{i=1}^{N}\left[y \cdot d^2 + (1 - y) \cdot \max(0, \text{margin} - d)^2\right] \quad (8)$$

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where y is the label of the sample pair, if the sample pair belongs to the same class, $y = 1$; if the sample pair belongs to a different class, $y = 0$. The margin is a hyperparameter that indicates the minimum distance between different sample pairs. N is the number of sample pairs, where $N = 1$. For positive samples, we aim to minimize the distance between them, while for negative samples, we want to ensure that the distance exceeds a specified margin.

D. Adaptive Spatial–Spectral Enhancement

When processing HSIs, spatial and spectral information are usually complementary, and considering either one alone may overlook some useful information. Moreover, in HTD, the spatial distribution of targets is usually sparse and localized, the targets are of diverse shapes and sizes, and the spectral features help to distinguish the targets from the background, while the complexity of the background increases the detection challenge. Therefore, TD algorithms must take into account the spatial and spectral characteristics of targets and be able to handle the complex relationship between targets and backgrounds. Considering the above issues, we propose an ASSE module, shown in Fig. 1, which contains two parts: spectral feature learning module (SFLM) and spatial constraint learning module (SCLM).

1) SFLM: Through a combination of pretraining and fine-tuning, GLSL not only maintains excellent spectral resolution but also demonstrates higher flexibility and adaptability in specific tasks. Specifically, after inputting the target prior spectra t and the spectra that need to be measured $x \in \mathbb{R}^{H^t \times W^t \times L^t}$ into the GLSL module to enhance the characteristics, the extracted features f_t and $f_{x_{i,j}}$ are used to compute the cosine similarity between the spectra to be measured and the target prior. Ultimately, a similarity map S is generated from these similarity values to efficiently capture and recognize subtle spectral differences

$$S(i, j) = \frac{f_t \cdot f_{x_{i,j}}}{\|f_t\| \left\| f_{x_{i,j}} \right\|} \tag{9}$$

where $S(i, j)$ denotes the similarity value at position $x_{i,j}$.

2) SCLM: In the learning phase of the ASSE, the SCLM module with multiple techniques including neighborhood operations and morphological operations is used to refine the target region and remove noise. In the neighborhood operation, by analyzing each target pixel and its neighborhood, the confidence level of these pixels is dynamically adjusted to better identify and preserve the real target region. Specifically, for the initial execution degree map S at position (i, j), the confidence value $S(i, j)$, a pixel point with a confidence level greater than r is set as the target pixel, the label is set to 1, and the value of r is set to 0.6. Then, the neighboring pixel points are analyzed with the target pixel as the center point. For example, to update the confidence level of each target pixel point based on a small range of 3×3 neighborhoods, we first calculate the sum $N_3(i, j)$ of the target pixel points in the 3×3 neighborhood of the point. If $N_3(i, j)$ is less than a set threshold value k_1, it means that the point is more likely to be an isolated noise point, and, therefore, the confidence level of the point needs to be reduced. Then, we use a decreasing exponential function q based on $N_3(i, j)$ to compute the weight w_3

$$w_3 = q(N_3(i, j)) = e^{-0.1 \cdot N_3(i,j)}. \tag{10}$$

For $N_3(i, j) < k_1$, the weights are applied to update the confidence map S

$$S'(i, j) = \max(S(i, j) - w_3, 0). \tag{11}$$

Given that the q-function is a decreasing function, the calculated weight w_3 has a larger value when the sum of labels in the neighborhood $N_3(i, j)$ is small. This indicates that the target pixel is more likely to be an isolated noise point and requires a greater reduction in its confidence level. On this basis, the confidence level of each target pixel point is updated in a large 7×7 neighborhood. If the sum of target pixel points in the neighborhood of the point $N_7(i, j)$ is greater than a set threshold k_2, it means that the target pixel neighborhood is likely to be a large pseudo-target, and, therefore, the confidence level of the point needs to be reduced. Then, an incremental exponential function $1 - q$ is used to compute the weights based on $N_7(i, j)$

$$w_7 = q(N_7(i, j)) = 1 - e^{-0.1 \cdot N_7(i,j)}. \tag{12}$$

For $N_7(i, j) > k_2$, the weights are applied to update the confidence map S'

$$S''(i, j) = \max\left(S'(i, j) - w_7, 0\right). \tag{13}$$

The incremental function is used because the larger $N_7(i, j)$ is the more likely it is to be a pseudo-target and more weights need to be reduced.

To optimize the boundaries of the target region and remove the noise from the edges while enhancing the target segmentation, we applied two morphological operations, erosion, and expansion, to S' as well as S'' after each neighborhood operation. Specifically, a 3×3 kernel is used for the first time to perform an erosion operation in the target region. This operation helps to remove smaller noise points without affecting the larger target structure. Then, a 7×7 kernel is used to perform an expansion operation on the eroded target region. The purpose of the expansion is to connect neighboring target regions so that the segmented target is more complete and coherent. Through the above operation, we can not only reduce false detection but also ensure that the boundary of the target region is more accurate. Finally, the TD result graph is derived from the updated similarity graph.

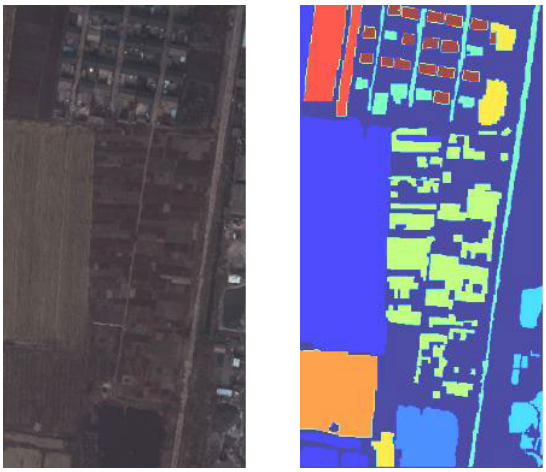
III. Experiments

A. Experimental Details

1) Data Description:

a) Source data: The source dataset is the Xuzhou [39] dataset, which was collected by an airborne HYSPEX hyperspectral camera over the Xuzhou peri-urban site in November 2014. This dataset consists of 500×260 pixels, with a very high spatial resolution of 0.73 m/pixel. The number of spectral bands used in the experiment was 436, after removing noisy bands ranging from 415 to 2508 nm. The scene is peri-urban

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(a) (b)

Fig. 3. Source data. (a) Pseudo-color image. (b) Ground truth.

and is characterized by nine categories, including crops, vegetation, man-made structures, and coal fields. The very high spatial resolution and the complex mixed categories make this dataset a challenging dataset for classification. Fig. 3 shows the pseudo-color image and the corresponding ground truth.

b) Airborne Visible/Infrared Imaging Spectrometer San Diego dataset [16]: The first dataset is the San Diego airport data, acquired by the airborne visible/infrared imaging spectrometer (AVIRIS) sensor, at the San Diego airport, CA, USA. This dataset contains three land covers, including the aircraft, the roof, and the runway of the airport, with a size of 120 × 120, a spectral resolution of 10 nm, and a spatial resolution of 3.5 m. The spectral length of this dataset varies from 370 to 2510 nm. Due to the water vapor absorption, several bands in the wavelength range of 1350–1420 nm and 1810–1940 nm are removed, and only 189 bands are preserved. The pseudo-color images and ground-truth maps are shown in Fig. 4(a).

c) Urban dataset [38]: The urban dataset is also popular in the field of HTD, which was captured by the AVIRIS sensor over the coast of Texas, USA. The spatial resolution of this dataset is about 17.2 m per pixel. The Urban dataset contains 100 × 100 pixels and there are 224 bands. Some bands suffer from noise, and after removing these low signal-to-noise bands, the dataset remains 204 bands. The detected target has a total of 67 pixels. Fig. 4(b) shows the pseudo-color image and ground truth.

d) Indian Pines dataset [40]: Indian Pines is a widely used hyperspectral dataset for classification and denoise. This dataset is also captured by the AVIRIS sensor over a single landscape in Indiana, US. The spatial size of this dataset is 145 × 145. For each pixel, the dataset contains 224 spectral reflectance bands which represent different portions of the electromagnetic spectra in the wavelength range 0.4–2.5 μm, while the original dataset contains some noised bands. After removing these bands, the dataset only contains 200 bands. Fig. 4(c) shows the pseudo-color image scene and the location of the target in this dataset.

e) Nuance-CRI Grass-Stone dataset [41]: The fourth dataset is the Nuance-CRI Grass-Stone dataset acquired by the Nuance CRI hyperspectral imager, a handheld imager with a spectral resolution of about 10 nm in 46 bands and a spectral range of 650–1100 nm. This image dataset contains 400 × 400 pixels and there are ten human-arranged stone targets of interest and several background features (e.g., bare soil and grass). The stone is selected as the target of interest in the experiment, and it occupies 0.007825 of the HSI. Due to the significant size differences in the stones in this data and the shadows cast on some stones due to slanting sunlight, accurate detection of the stones has become quite challenging. Fig. 4(d) shows the pseudo-color image scene and the location of the target in this dataset.

2) Comparison Methods: To verify the effectiveness of the proposed HTD algorithm, we selected seven advanced HTD algorithms as comparison algorithms, including two classical HTD methods: CEM [6] and HCEM [8], two representation-based methods: CSCR [14] and DSC [42], and three latest deep learning-based methods: MLSN [43], LCNN-CD [33], and HTD-IRN [31]. In particular, both the proposed and the comparison methods use the same prior target spectra for HTD.

3) Evaluation Criterion: To analyze the performance of the proposed SelfMTL detection, three qualitative and quantitative evaluation metrics are used: receiver operating characteristic (ROC) curve, area under the curve (AUC) value, and separability plots.

The ROC curve is a common tool for evaluating the performance of binary classification models. A 3-D ROC (3D ROC) curve is used to measure detector performance. The 3D ROC curve can be considered as a function of the detection probability D, the false alarm probability F, and the threshold τ. With a given τ, D and F can be calculated by

$$D_\tau = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \quad F_\tau = \frac{\mathrm{FP}}{\mathrm{FP}+\mathrm{TN}} \tag{14}$$

where TP denotes the number of pixels that are correctly detected as targets at a given threshold τ, FN represents the number of pixels that incorrectly detect targets as backgrounds at a given threshold τ, FP denotes the number of pixels that incorrectly detect backgrounds as targets at a given threshold τ, and TN denotes the number of pixels that are correctly detected as backgrounds at a given threshold τ. The plotted 2D ROC images show the false-positive rate on the horizontal axis and the true-positive rate on the vertical axis. The horizontal axis represents the proportion of negative samples that were misclassified as positive by the classifier, while the vertical axis represents the proportion of positive samples that were correctly classified as positive. In general, the closer the ROC curve is to the upper left corner, the better the performance of the classifier is, while the closer the curve is to the diagonal, the worse the performance of the classifier is.

A set of AUC variants is derived from variables D, F, and τ. The classical $\mathrm{AUC}_{(F,D)}$ measures the area under the ROC curve formed by the true F and D. A higher value of $\mathrm{AUC}_{(F,D)}$ indicates better detection effectiveness. $\mathrm{AUC}_{(\tau,F)}$ measures the

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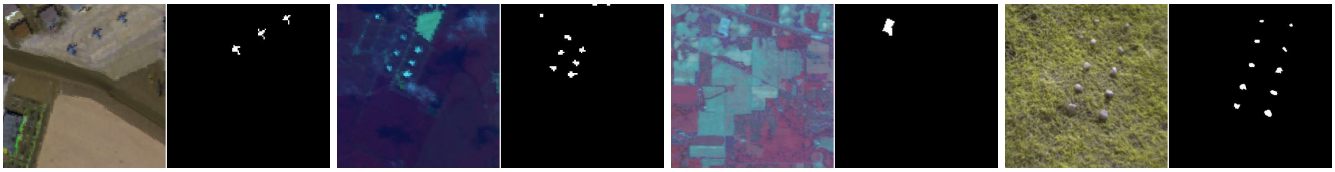


Fig. 4. Dataset image scenes and ground truth. (a) San Diego. (b) Urban. (c) Indian Pines. (d) Nuance-CRI.

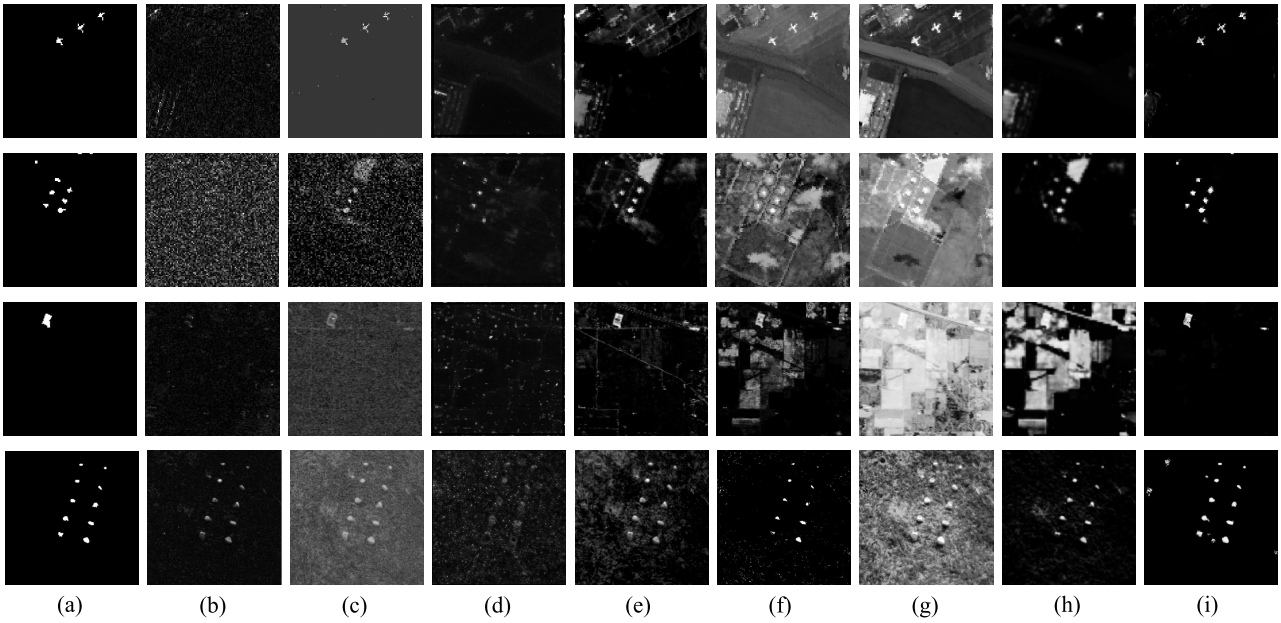


Fig. 5. Detection maps for different methods. (a) Ground truth. (b) CEM. (c) HCEM. (d) CSCR. (e) DSC. (f) MLSN. (g) LCNN-CD. (h) HTD-IRN. (i) SelfMTL.

area under the ROC curve of the threshold τ versus F. A lower value $\mathrm{AUC}_{(\tau,F)}$ signifies superior background suppression performance. Similarly, $\mathrm{AUC}_{(\tau,D)}$ measures the area under the ROC curve of the threshold τ versus the D. A higher value $\mathrm{AUC}_{(\tau,D)}$ indicates a better TD performance. Based on these criteria, three additional AUC variants have been designed to comprehensively evaluate detector performance

$$\mathrm{AUC}_{\mathrm{TD}} = \mathrm{AUC}_{(F,D)} + \mathrm{AUC}_{(\tau,D)} \tag{15}$$

$\mathrm{AUC}_{\mathrm{TD}}$ further demonstrates the overall TD capability

$$\mathrm{AUC}_{\mathrm{BS}} = \mathrm{AUC}_{(F,D)} - \mathrm{AUC}_{(\tau,F)} \tag{16}$$

$\mathrm{AUC}_{\mathrm{BS}}$ further demonstrates background inhibition

$$\mathrm{AUC}_{\mathrm{ODP}} = 1 + \mathrm{AUC}_{(\tau,D)} - \mathrm{AUC}_{(\tau,F)} \tag{17}$$

$\mathrm{AUC}_{\mathrm{ODP}}$ further demonstrates the combined performance of TD and background suppression.

This comprehensive set of metrics allows for a detailed assessment of the performance of different detectors in both TD and background suppression tasks.

The separability plot [44] is a statistics-based method to visualize the separability between positive and negative samples in binary hypothesis testing problems. This graphical representation relies on box plots of different colors to represent the data distributions of the positive and negative samples separately. To observe the position and range of the box plot, the distinguishability between the targets and the backgrounds is intuitively revealed. The median is located in the center of the box plot, the quartiles define the upper and lower boundaries of the box plot, and the whiskers show the range of data excluding outliers. This graph not only visually demonstrates the distribution of data, but also directly reflects the level of separability between the targets and backgrounds by comparing the overlap of box plots.

4) Experimental Settings: For the proposed SelfMTL method, the learning rate of the pretrained GLSL network is set to 1e-3. The number of epochs and batch size are set to 45 and 64, respectively. The learning rate and the number of epochs for the fine-tuned GLSL network of the small sample are set to 1e-2 and 10, respectively, and the Adam optimizer is used. All experiments were carried out using an NVIDIA GeForce RTX 3090 graphics processor (GPU) and 16 GB of random access memory (RAM).

B. Detection Performance

This section analyzes the detection performance of the proposed SelfMTL method compared to other detection methods. We evaluated the effectiveness of transferring the pretrained model to various TD tasks with four different datasets.

Fig. 5 visualizes the detection maps of SelfMTL and the other comparison methods. It can be seen that SelfMTL strikes a satisfactory balance between overall accuracy and

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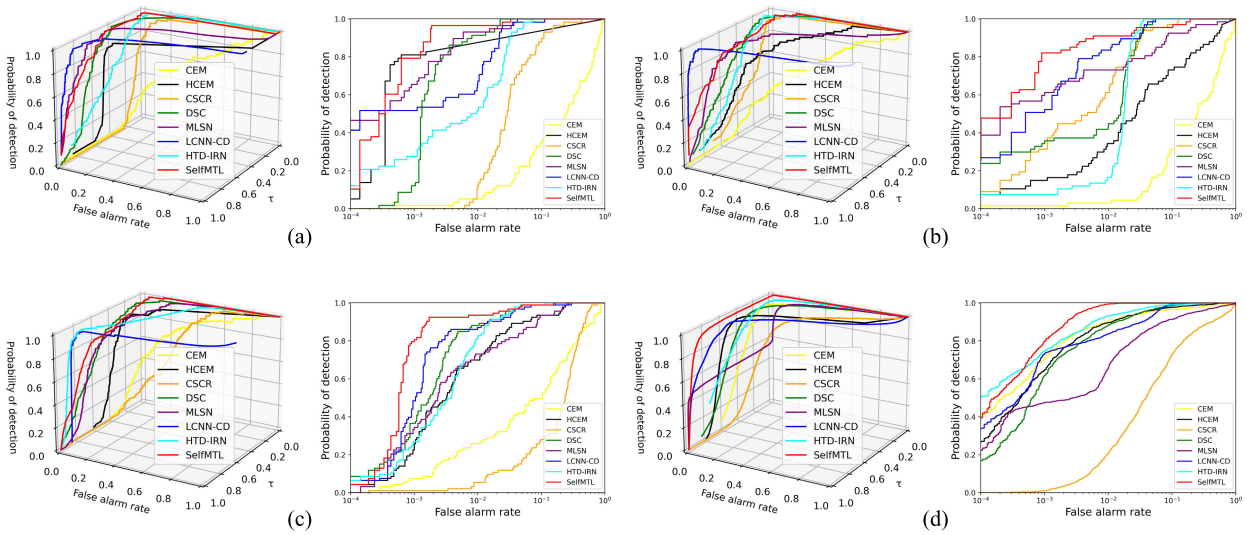


Fig. 6. ROC curves for all the detectors on the four datasets. (a) San Diego. (b) Urban. (c) Indian Pines. (d) Nuance-CRI.

background suppression compared to its competitors. For example, there are large near-target background regions interfering in the lower left corner of the San Diego dataset, the right edge of the Indian Pines dataset, and the upper right of the targets in the Urban dataset, and thus many comparative methods, including DSC, MLSN, and LCNN-CD, are unable to suppress the background effectively. CEM and HCEM can only detect part of the targets on the four datasets and are unable to effectively suppress the background. CSCR and HTD-IRN perform better in suppressing the background interference on the four datasets, but only detect some targets and are poor in detecting the target locations. DSC and MLSN can effectively detect most of the targets, but have difficulty in suppressing the background well, and occasionally misclassify some of the background as a target, especially on the San Diego dataset and the Urban dataset. SelfMTL shows strong background suppression on all four datasets. Overall, our method shows a more stable detection rate as well as excellent background suppression on all four datasets. The main reason may be that SelfMTL employs a global–local feature extraction module, which can separate the spectral feature variability effectively.

The 3D and derived 2D ROC curves for the four datasets are shown in Fig. 6, where the red curve represents the proposed SelfMTL. It can be seen that SelfMTL consistently achieves the best 3D ROC profile on all four datasets, with a higher detection probability and a lower false alarm rate. Furthermore, we find that the 2D ROC curves of (D, F) of SelfMTL for the four datasets are located almost in the upper left corner, indicating its promising and robust performance. For the Nuance-CRI dataset, the ROC curve for the SelfMTL detector is slightly lower when the false alarm rate is less than 0.001. This may be attributed to the presence of an extreme imbalance between the target sample and the background sample categories in the Nuance-CRI dataset.

The AUC values for all the methods on the four datasets are given in Table I, with $\text{AUC}_{(F,D)}$ reflecting detection accuracy and $\text{AUC}_{(\tau,F)}$ responding to background suppression accuracy, and $\text{AUC}_{(\tau,D)}$ responding to the detectability of the target. The closer the values of $\text{AUC}_{(F,D)}$ and $\text{AUC}_{(\tau,D)}$ are to 1, the better the effect is, and the closer the value of $\text{AUC}_{(\tau,F)}$ is to 0, the better the effect is. The optimal results are shown in bold, and the suboptimal results are underlined. It is clear that the SelfMTL produces the most accurate and reliable background suppression on the $\text{AUC}_{(\tau,F)}$ metric compared to its competitors. Especially on the San Diego and Urban datasets, where background is more difficult to suppress, it far outperforms the other comparison methods. In particular, all six metrics are optimized on the Nuance-CRI dataset, while more than half of the metrics are optimized on the other three datasets. Overall, consistent with the above analysis, SelfMTL obtains the most satisfactory results on the four datasets, effectively and efficiently.

The separability plots of each method on the four datasets are shown in Fig. 7 to intuitively validate the effectiveness of our method in separating targets from backgrounds. For almost all the datasets, compared to the other methods, our method consistently exhibits larger gaps between the magenta boxes (representing targets) and the black boxes (representing backgrounds), indicating a better target- and background-separability ability. Additionally, the central lines of the magenta boxes for our method are generally positioned higher, suggesting that SelfMTL can better highlight targets. On the four datasets, SelfMTL suppresses the detected values of background pixels in the detection results to a very low level, showing excellent background suppression. Even on the Urban dataset, where the background of the detection scene is complex, the target and the background in the SelfMTL detection results still have good separability. Comparative experiments show that the GLSL feature extraction module obtained through pretraining can effectively enhance the spectral difference discrimination ability, while the joint spatial–spectral constraint learning can effectively enhance the background suppression ability.

C. Parameter Analysis

This section analyzes two crucial parameters of the proposed SelfMTL: the margin threshold parameter,

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TABLE I
AUC VALUES ON THE FOUR DATASETS

Datasets	Metric	Methods							
		CEM	HCEM	CSCR	DSC	MLSN	LCNN-CD	HTD-IRN	SelfMTL
San Diego	$AUC_{(F,D)}$	0.65977	0.96549	0.95085	0.99667	0.99688	0.99139	0.98597	**0.99861**
	$AUC_{(\tau,F)}$	0.06297	0.23936	0.05889	0.05761	0.32283	0.13162	0.02820	**0.00726**
	$AUC_{(\tau,D)}$	0.11104	0.60158	0.15334	0.69399	**0.86025**	0.49511	0.50429	0.79240
	AUC_{TD}	0.77081	1.56707	1.10419	1.69067	**1.85713**	1.48651	1.49026	1.79101
	AUC_{BS}	0.59680	0.72613	0.89197	0.93906	0.67405	0.85978	0.95777	**0.99134**
	AUC_{ODP}	1.04807	1.36222	1.09446	1.63638	1.53742	1.36349	1.47609	**1.78514**
Urban	$AUC_{(F,D)}$	0.65786	0.88095	0.98625	0.98659	0.96065	**0.99423**	0.98342	0.99143
	$AUC_{(\tau,F)}$	0.19479	0.11277	0.05208	0.04744	0.31573	0.14309	0.04701	**0.00213**
	$AUC_{(\tau,D)}$	0.29000	0.44154	0.31050	0.64496	**0.76622**	0.26998	0.53879	0.73934
	AUC_{TD}	0.94786	1.32249	1.29675	1.63156	1.72688	1.26421	1.52222	**1.73076**
	AUC_{BS}	0.46306	0.76819	0.93417	0.93915	0.64492	0.85114	0.93641	**0.98929**
	AUC_{ODP}	1.09520	1.32878	1.25842	1.59752	1.45050	1.12688	1.49178	**1.73721**
Indian Pines	$AUC_{(F,D)}$	0.80510	0.97736	0.75776	0.99369	0.97499	0.99196	0.99252	**0.99589**
	$AUC_{(\tau,F)}$	0.04567	0.15943	0.05558	0.05164	0.09184	0.19767	0.25704	**0.00289**
	$AUC_{(\tau,D)}$	0.12338	0.39949	0.10003	0.66316	0.64208	0.27067	**0.91469**	0.70678
	AUC_{TD}	0.92848	1.37686	0.85778	1.65686	1.61706	1.26264	**1.90722**	1.70267
	AUC_{BS}	0.75943	0.81794	0.70218	0.94206	0.88314	0.79429	0.73549	**0.99299**
	AUC_{ODP}	1.07771	1.24007	1.04445	1.61152	1.55024	1.07300	1.65766	**1.70388**
Nuance-CRI	$AUC_{(F,D)}$	0.98219	0.98935	0.87118	0.99312	0.95911	0.99156	0.99509	**0.99832**
	$AUC_{(\tau,F)}$	0.05070	0.30965	0.09146	0.09426	0.00488	0.07693	0.02847	**0.00466**
	$AUC_{(\tau,D)}$	0.39375	0.64463	0.24455	0.57452	0.52408	0.16577	0.56428	**0.85917**
	AUC_{TD}	1.37594	1.63398	1.11573	1.56764	1.48319	1.15734	1.55937	**1.85749**
	AUC_{BS}	0.93148	0.67969	0.77972	0.89885	0.95422	0.91464	0.96662	**0.99366**
	AUC_{ODP}	1.34305	1.33498	1.15308	1.48026	1.51919	1.08885	1.53581	**1.85451**

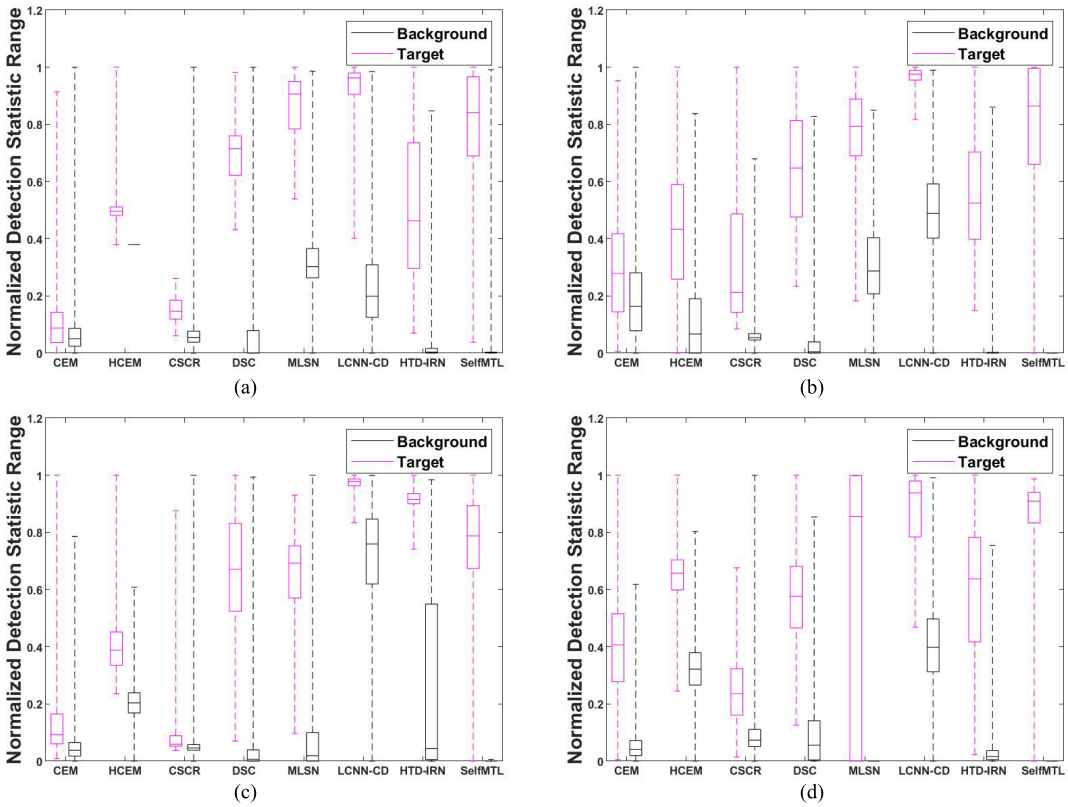


Fig. 7. Maps of target background separation for all detection methods on the four datasets: (a) San Diego. (b) Urban. (c) Indian Pines. (d) Nuance-CRI.

which controls the degree of target–background separation in the contrast loss function during fine-tuning experiments, and the dilation kernel size parameter in the joint spectral–spatial constraint learning module.

In the small-sample learning phase, we used a contrast loss function for optimization, and since there are only two samples for training, it is important to effectively balance the differences between the target and the background. Among them,

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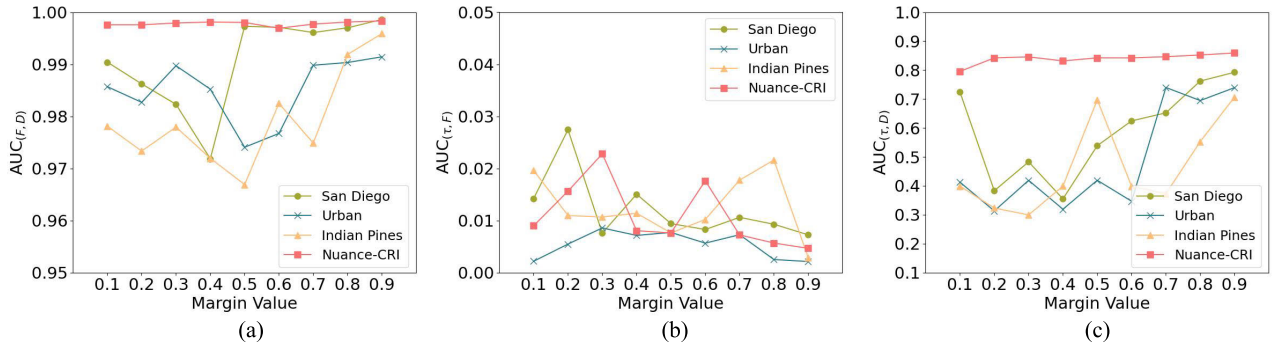


Fig. 8. Parameter analysis on four datasets for margin value. (a) $\mathrm{AUC}_{(F,D)}$. (b) $\mathrm{AUC}_{(\tau,F)}$. (c) $\mathrm{AUC}_{(\tau,D)}$.

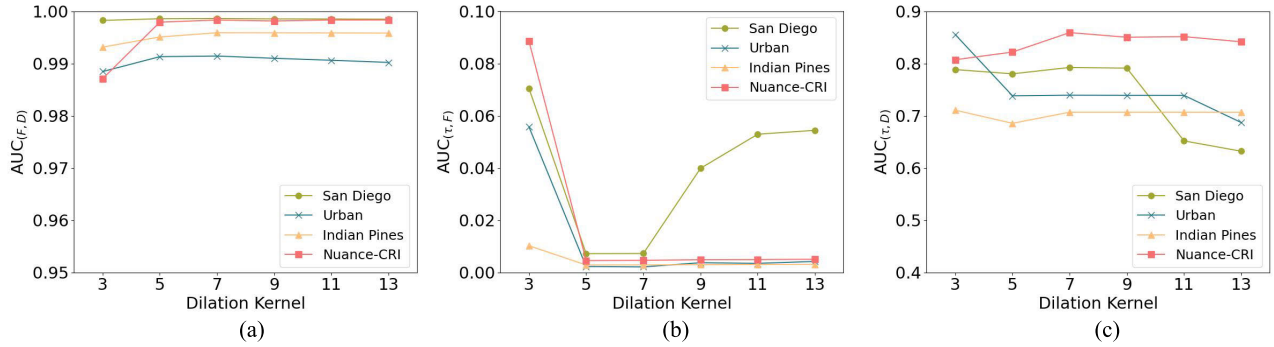


Fig. 9. Parameter analysis on four datasets for dilation kernel. (a) $\mathrm{AUC}_{(F,D)}$. (b) $\mathrm{AUC}_{(\tau,F)}$. (c) $\mathrm{AUC}_{(\tau,D)}$.

the margin threshold parameter sets a clear boundary to distinguish the distance between the target features and the background features, and choosing an appropriate threshold can ensure that the model not only recognizes the target but also effectively suppresses the background noise. Thus, based on the similarity between positive and negative sample pairs, we set the candidate parameter set ([0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]). To mitigate the effect of the random factor, we performed five consecutive runs and the average $\mathrm{AUC}_{(F,D)}$, $\mathrm{AUC}_{(\tau,F)}$, and $\mathrm{AUC}_{(\tau,D)}$ on the four datasets are shown in Fig. 8. When the value of the margin threshold is taken as small, the results of all four datasets do not work well. This may be because the model focuses more on the distance between pairs of positive samples and the background suppression is weaker when the margin threshold is small. The Nuance-CRI dataset shows less fluctuation over the entire range of the margin threshold, possibly due to the relatively simple background composition of the Nuance-CRI dataset. In general, smaller margin thresholds may cause the model to have difficulty in effectively distinguishing between the target and the background, which, in turn, affects overall detection performance. Considering the overall analysis of the four datasets, we chose 0.9 as the margin threshold of the contrast loss function.

In the learning phase of the ASSE, various techniques, including neighborhood operations and morphological operations, are used to refine the target region and remove noise. Specifically, erosion and dilation operations are used to optimize the boundaries of the target region. Among them, the erosion operation is fixed using a 3×3 kernel, which helps to remove smaller noise points without affecting the larger target structure. For the dilation operation, given the uncertainty of the target size, the size of the kernel needs to be further determined to ensure that it connects neighboring target regions without overextending the boundaries and introducing additional noise. Thus, we set the set of candidate parameters for the size of the kernel ([3, 5, 7, 9, 11, 13]). To mitigate the effect of the random factor, we performed five consecutive runs and the average $\mathrm{AUC}_{(F,D)}$, $\mathrm{AUC}_{(\tau,F)}$, and $\mathrm{AUC}_{(\tau,D)}$ on the four datasets are shown in Fig. 9. From the three types of results on the four datasets, it is clear that a kernel that is too small cannot effectively distinguish the background noise from the target region, especially when the background noise is in close proximity to the target pixels. This can lead to background noise being mistaken as part of the target, which reduces the ability of background suppression. An oversized kernel increases the computational burden, while also changing the original shape of the target. Therefore, considering the overall analysis of the four datasets, we chose 7 as the value of the dilated kernel.

D. Ablation Study

To show the effectiveness of the proposed modules more clearly, we performed ablation experiments on each of the four datasets, which includes the GLSL and SCLM modules. Meanwhile, to study the properties of a total of three GLSL modules in more detail, we added modules step by step and designed five experiments. The optimal results are shown in bold, and the suboptimal results are underlined. From the

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TABLE II
COMPARISONS ABLATION OF EACH MODULE ON THREE DATASETS

Dataset	Model	(1)	(2)	(3)	(4)	(5)
	LV	✓	×	×	✓	✓
	GV	×	✓	×	✓	✓
	SFFM	×	×	✓	✓	✓
	SCLM	×	×	×	×	✓
San Diego	$AUC_{(F,D)}$	0.94957	0.96519	0.97619	0.99292	**0.99861**
	$AUC_{(\tau,F)}$	0.67937	0.00741	0.68855	0.00739	**0.00726**
	$AUC_{(\tau,D)}$	0.76087	0.75367	0.74644	0.76051	**0.79240**
Urban	$AUC_{(F,D)}$	0.91441	0.98013	0.98222	0.98627	**0.99143**
	$AUC_{(\tau,F)}$	0.74011	0.01312	0.81181	0.01575	**0.00213**
	$AUC_{(\tau,D)}$	0.68236	0.64397	0.68141	0.68371	**0.73934**
Indian Pines	$AUC_{(F,D)}$	0.97346	0.98203	0.98499	0.98891	**0.99586**
	$AUC_{(\tau,F)}$	0.10649	0.03293	0.40634	0.01856	**0.00294**
	$AUC_{(\tau,D)}$	0.69837	0.53827	0.25095	0.69686	**0.70678**
Nuance-CRI	$AUC_{(F,D)}$	0.93276	0.98025	0.98942	0.99693	**0.99832**
	$AUC_{(\tau,F)}$	0.76846	0.01634	0.45823	0.01251	**0.00466**
	$AUC_{(\tau,D)}$	0.79485	0.66532	0.67254	0.85171	**0.85917**

experimental results shown in Table II, it can be seen that with the gradual addition of the proposed modules, the accuracy is generally improved with respect to the previous model and the complete model has the highest accuracy. This shows the effectiveness of the individual modules.

a) Effectiveness of GLSL: This section analyzes the necessity and effectiveness of the GLSL module. The GLSL module is designed to acquire global and local spectral information to effectively learn the similarities between similar spectra and the differences between different classes of spectra. We find that the overall accuracy of TD is maintained at a high level when the SFFM module is used individually on all datasets in the experiment (1). Meanwhile, the San Diego dataset, as well as the Indian Pines dataset, have $\text{AUC}_{(\tau,D)}$ metrics that perform second only to the full model in Experiment (2), which shows the contribution of the LV module to highlighting the target, which becomes clearer and more prominent as it learns the features at different scales through double-branching and multiscale convolution. It is easy to see the contribution of the GV module in experiment (3), which learns a more accurate background distribution by capturing the remote contextual relationship between channels, effectively suppressing background pixels. Then, the indicator $\text{AUC}_{(\tau,F)}$ performs well on all the datasets. In experiment (4), combining the advantages of these three modules, the overall accuracy, the background suppression system, and the TD rate are significantly improved.

b) Effectiveness of SCLM: This section analyzes the necessity and effectiveness of the SCLM module. The SCLM module is designed to combine spectral features with spatial constraints to achieve joint spatial–spectral learning. Comparing experiments (4) and (5), it can be observed that the proposed SCLM achieves the best overall detection accuracy, background suppression, and TD rate across all the datasets. This may be attributed to the SCLM module's ability to effectively integrate target location information, significantly reducing scattered background noise in the target domain and suppressing large areas of false positives. On average, the results indicate that, by balancing spectral and spatial information, SCLM outperforms methods based solely on spectral features, demonstrating good generalization across different datasets.

IV. CONCLUSION

This article proposed a novel HTD method named SelfMTL. Specifically, to solve the problem of limited training samples for training deep neural networks, as well as the weak generalization ability and adaptation to new tasks of deep learning-based HTD algorithms. SelfMTL proposes to learn a pretrained network by GLSL, which is effectively migrated to each different TD task. In addition, to solve the problem of neglecting spatial information in previous methods, SelfMTL designs a joint spatial–spectral constraint learning module to refine the target region and remove the noise to further improve the background suppression capability. Extensive experimental results on four challenging hyperspectral datasets validate the superiority of the SelfMTL method over several state-of-the-art approaches.

In future work, we will look more deeply into how spectral differences can be exploited to achieve further suppression of the background, find more effective cores, and optimize the network structure.

REFERENCES

[1] W. Dong et al., "Deep spatial–spectral joint-sparse prior encoding network for hyperspectral target detection," *IEEE Trans. Cybern.*, vol. 54, no. 12, pp. 7780–7792, Dec. 2024.

[2] D. Zhu, P. Zhong, B. Du, and L. Zhang, "Attention-based sparse and collaborative spectral abundance learning for hyperspectral subpixel target detection," *Neural Netw.*, vol. 178, Oct. 2024, Art. no. 106416.

[3] H. Wang, Z. Zhong, F. Lei, J. Peng, and S. Yue, "Bio-inspired small target motion detection with spatio-temporal feedback in natural scenes," *IEEE Trans. Image Process.*, vol. 33, pp. 451–465, 2024.

[4] L. Caruccio, S. Cirillo, G. Polese, G. Solimando, S. Sundaramurthy, and G. Tortora, "Can ChatGPT provide intelligent diagnoses? A comparative study between predictive models and ChatGPT to define a new medical diagnostic bot," *Expert Syst. Appl.*, vol. 235, Jan. 2024, Art. no. 121186.

[5] V. Bayarri, A. Prada, F. García, C. De Las Heras, and P. Fatás, "Remote sensing and environmental monitoring analysis of pigment migrations in cave of Altamira's prehistoric paintings," *Remote Sens.*, vol. 16, no. 12, p. 2099, Jun. 2024.

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[6] C.-I. Chang and D. Heinz, "Constrained subpixel target detection for remotely sensed imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1144–1159, May 2000.

[7] C.-I. Chang, "Orthogonal subspace projection (OSP) revisited: A comprehensive study and analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 3, pp. 502–518, Mar. 2005.

[8] Z. Zou and Z. Shi, "Hierarchical suppression method for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 1, pp. 330–342, Jan. 2015.

[9] R. Zhao, Z. Shi, Z. Zou, and Z. Zhang, "Ensemble-based cascaded constrained energy minimization for hyperspectral target detection," *Remote Sens.*, vol. 11, no. 11, p. 1310, Jun. 2019.

[10] H. Kwon and N. M. Nasrabadi, "Kernel orthogonal subspace projection for hyperspectral signal classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 43, no. 12, pp. 2952–2962, Dec. 2005.

[11] X. Jiao and C.-I. Chang, "Kernel-based constrained energy minimization (K-CEM)," *Proc. SPIE*, vol. 6966, pp. 523–533, Apr. 2008.

[12] H. Kwon and N. M. Nasrabadi, "Kernel matched subspace detectors for hyperspectral target detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 28, no. 2, pp. 178–194, Feb. 2006.

[13] Y. Chen, N. M. Nasrabadi, and T. D. Tran, "Sparse representation for target detection in hyperspectral imagery," *IEEE J. Sel. Topics Signal Process.*, vol. 5, no. 3, pp. 629–640, Jun. 2011.

[14] W. Li, Q. Du, and B. Zhang, "Combined sparse and collaborative representation for hyperspectral target detection," *Pattern Recognit.*, vol. 48, no. 12, pp. 3904–3916, Dec. 2015.

[15] X. Lu, W. Zhang, and X. Li, "A hybrid sparsity and distance-based discrimination detector for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 3, pp. 1704–1717, Mar. 2018.

[16] D. Zhu, B. Du, and L. Zhang, "Learning single spectral abundance for hyperspectral subpixel target detection," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 10134–10144, Jul. 2024.

[17] Y. Dong, W. Shi, B. Du, X. Hu, and L. Zhang, "Asymmetric weighted logistic metric learning for hyperspectral target detection," *IEEE Trans. Cybern.*, vol. 52, no. 10, pp. 11093–11106, Oct. 2022.

[18] X. Sun et al., "Target detection through tree-structured encoding for hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 5, pp. 4233–4249, May 2021.

[19] B. Yang, Y. Mao, L. Liu, L. Fang, and X. Liu, "Change representation and extraction in stripes: Rethinking unsupervised hyperspectral image change detection with an untrained network," *IEEE Trans. Image Process.*, vol. 33, pp. 5098–5113, 2024.

[20] M. Hu, C. Wu, B. Du, and L. Zhang, "Binary change guided hyperspectral multiclass change detection," *IEEE Trans. Image Process.*, vol. 32, pp. 791–806, 2023.

[21] J. Bai, W. Shi, Z. Xiao, T. A. A. Ali, F. Ye, and L. Jiao, "Achieving better category separability for hyperspectral image classification: A spatial–spectral approach," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 7, pp. 9621–9635, Jul. 2024.

[22] Y. Su, L. Gao, M. Jiang, A. Plaza, X. Sun, and B. Zhang, "NSCKL: Normalized spectral clustering with kernel-based learning for semisupervised hyperspectral image classification," *IEEE Trans. Cybern.*, vol. 53, no. 10, pp. 6649–6662, Oct. 2023.

[23] Y. Fang, L. Sun, Y. Zheng, and Z. Wu, "Deformable convolution-enhanced hierarchical transformer with spectral–spatial cluster attention for hyperspectral image classification," *IEEE Trans. Image Process.*, vol. 34, pp. 701–716, 2025.

[24] R. A. Borsoi, T. Imbiriba, and P. Closas, "Dynamical hyperspectral unmixing with variational recurrent neural networks," *IEEE Trans. Image Process.*, vol. 32, pp. 2279–2294, 2023.

[25] M. Wang, Q. Wang, D. Hong, S. K. Roy, and J. Chanussot, "Learning tensor low-rank representation for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 53, no. 1, pp. 679–691, Jan. 2023.

[26] S. Sun, J. Liu, and W. Li, "Spatial invariant tensor self-representation model for hyperspectral anomaly detection," *IEEE Trans. Cybern.*, vol. 54, no. 5, pp. 3120–3131, May 2024.

[27] J. Ma, W. Xie, and Y. Li, "Exploring hyperspectral anomaly detection with human vision: A small target aware detector," *Neural Netw.*, vol. 184, Apr. 2025, Art. no. 107036.

[28] G. Zhang, S. Zhao, W. Li, Q. Du, Q. Ran, and R. Tao, "HTD-Net: A deep convolutional neural network for target detection in hyperspectral imagery," *Remote Sens.*, vol. 12, no. 9, p. 1489, May 2020.

[29] D. Zhu, B. Du, and L. Zhang, "Two-stream convolutional networks for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6907–6921, Aug. 2021.

[30] Y. Gao, Y. Feng, and X. Yu, "Hyperspectral target detection with an auxiliary generative adversarial network," *Remote Sens.*, vol. 13, no. 21, p. 4454, Nov. 2021.

[31] D. Shen, X. Ma, W. Kong, J. Liu, J. Wang, and H. Wang, "Hyperspectral target detection based on interpretable representation network," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–16, 2023, Art no. 5519416.

[32] D. Zhu, B. Du, Y. Dong, and L. Zhang, "Target detection with spatial–spectral adaptive sample generation and deep metric learning for hyperspectral imagery," *IEEE Trans. Multimedia*, vol. 25, pp. 6538–6550, 2023.

[33] Y. Dong, X. Dai, Y. Zhang, and B. Du, "A lightweight convolutional neural network based on joint correlation distance constraints and density peak clustering for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–14, 2023, Art no. 5516514.

[34] Y. Shi, J. Li, Y. Li, and Q. Du, "Sensor-independent hyperspectral target detection with semisupervised domain adaptive few-shot learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 8, pp. 6894–6906, Aug. 2021.

[35] Y. Wang, X. Chen, E. Zhao, and M. Song, "Self-supervised spectral-level contrastive learning for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–15, 2023, Art no. 5510515.

[36] X. Zhang et al., "Self-supervised learning with deep clustering for target detection in hyperspectral images with insufficient spectral variation prior," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 122, Aug. 2023, Art. no. 103405.

[37] X. Chen, Y. Zhang, Y. Dong, and B. Du, "Generative self-supervised learning with spectral–spatial masking for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1–13, 2024, Art no. 5522713.

[38] J. Jiao, Z. Gong, and P. Zhong, "Triplet spectralwise transformer network for hyperspectral target detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, pp. 1–17, 2023, Art no. 5519817.

[39] T. Dutta, S. Dey, S. Bhattacharyya, and S. Mukhopadhyay, "Quantum fractional order darwinian particle swarm optimization for hyperspectral multi-level image thresholding," *Appl. Soft Comput.*, vol. 113, 2021, Art. no. 107976.

[40] J. Shi, T. Wu, A. K. Qin, T. Shao, Y. Lei, and G. Jeon, "Deep-growing neural network with manifold constraints for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 1, pp. 210–221, Jan. 2025.

[41] Y. Zhang, B. Du, L. Zhang, and S. Wang, "A low-rank and sparse matrix decomposition-based Mahalanobis distance method for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 3, pp. 1376–1389, Mar. 2016.

[42] D. Shen, X. Ma, H. Wang, and J. Liu, "A dual sparsity constrained approach for hyperspectral target detection," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2022, pp. 1963–1966.

[43] Y. Wang, X. Chen, F. Wang, M. Song, and C. Yu, "Meta-learning based hyperspectral target detection using Siamese network," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–13, 2022, Art no. 5527913.

[44] T. Guo, F. Luo, J. Guo, Y. Duan, X. Huang, and G. Shi, "Hyperspectral target detection with target prior augmentation and background suppression-based multidetector fusion," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 17, pp. 1765–1780, 2024.

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